

Wave and Optics

Part-1



Dr. Ram Prakash
Professor
Department of Physics
Indian Institute of Technology (IIT), Jodhpur

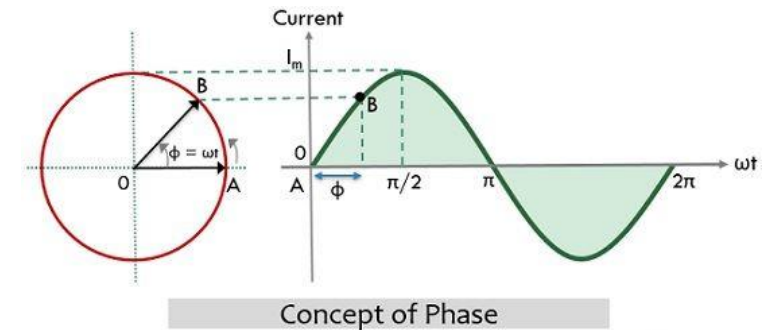
Email: ramprakash@iitj.ac.in

Interference

When two waves superimpose one obtain an intensity distribution, which is known as interference pattern and the phenomena which causes this variation is known as interference.

Consider superposition of two non-identical (having time independent constant amps & of different phases) time dependent sinusoidal transverse waves of same frequency (ω) :

$$x_1(t) = a_1 \cos(\omega t + \varphi_1) = a_1 \cos \omega t \cos \varphi_1 - a_1 \sin \omega t \sin \varphi_1$$
$$x_2(t) = a_2 \cos(\omega t + \varphi_2) = a_2 \cos \omega t \cos \varphi_2 - a_2 \sin \omega t \sin \varphi_2$$



Resulting displacement due to superposition of the two will be given by

$$x(t) = x_1(t) + x_2(t) = \underbrace{(a_1 \cos \varphi_1 + a_2 \cos \varphi_2)}_{\equiv a \cos \varphi} \cos \omega t - \underbrace{(a_1 \sin \varphi_1 + a_2 \sin \varphi_2)}_{\equiv a \sin \varphi} \sin \omega t$$

$$\Rightarrow x(t) = a \cos \varphi \cos \omega t - a \sin \varphi \sin \omega t$$
$$\Rightarrow x(t) = a \cos(\omega t + \varphi)$$

where

$$\begin{aligned} a \cos \varphi &= (a_1 \cos \varphi_1 + a_2 \cos \varphi_2) \\ a \sin \varphi &= (a_1 \sin \varphi_1 + a_2 \sin \varphi_2) \dots \dots (1) \end{aligned}$$

By squaring and adding

$$\begin{aligned} a^2(\cos^2 \varphi + \sin^2 \varphi) &= a_1^2[\cos^2 \varphi_1 + \sin^2 \varphi_1] + a_2^2[\cos^2 \varphi_2 + \sin^2 \varphi_2] \\ &\quad + 2 a_1 a_2 [\cos \varphi_1 \cos \varphi_2 + \sin \varphi_1 \sin \varphi_2] \\ \Rightarrow a^2 &= a_1^2 + a_2^2 + 2 a_1 a_2 [\cos \varphi_1 \cos \varphi_2 + \sin \varphi_1 \sin \varphi_2] \end{aligned}$$

Taking square root on both sides, we get

$$a = [a_1^2 + a_2^2 + 2a_1 a_2 \cos(\varphi_1 - \varphi_2)]^{1/2}$$

We also get from eq. (1)

$$\tan \varphi = \frac{a_1 \sin \varphi_1 + a_2 \sin \varphi_2}{a_1 \cos \varphi_1 + a_2 \cos \varphi_2} \Rightarrow \varphi = \tan^{-1} \left[\frac{a_1 \sin \varphi_1 + a_2 \sin \varphi_2}{a_1 \cos \varphi_1 + a_2 \cos \varphi_2} \right]$$

Thus, a and φ are now known

From

$$a = [a_1^2 + a_2^2 + 2a_1a_2 \cos(\varphi_1 - \varphi_2)]^{1/2} = [a_1^2 + a_2^2 + 2a_1a_2 \cos(\Delta\varphi)]^{1/2}$$

$$\text{where } \Delta\varphi = \varphi_1 - \varphi_2$$

If $\Delta\varphi = 0, 2\pi, 4\pi, \dots \equiv 2m\pi ; m = 0, 1, 2, 3, \dots$

$$a = [a_1^2 + a_2^2 + 2a_1a_2]^{1/2} \quad \text{for} \quad \Delta\varphi = 2m\pi ; m = 0, 1, 2, 3, \dots$$

$$= [(a_1 + a_2)^2]^{1/2} = a_1 + a_2$$

$\Rightarrow$ Two waves are in phase

If the two waves are in phase, two amplitudes add up to form the resultant amplitude :

$$a = a_1 + a_2$$

This is called constructive interference

If $\Delta\varphi = \varphi_1 \sim \varphi_2$

$$= (2p + 1)\pi; p = 0, 1, 2, 3, \dots \Rightarrow \cos \Delta\varphi = -1$$

$$\Rightarrow a = [(a_1 \sim a_2)^2]^{1/2}$$

$$\Rightarrow a = a_1 \sim a_2$$

$\Rightarrow$ Two waves are out of phase, the resultant amplitude is the difference between the two amplitudes to form destructive interference

Thus if ,

$$a_1 = a_2 (= U)$$

Then, in case of **constructive interference** resultant amplitude will be

$$a = 2U$$

In case of **destructive interference**, resultant amplitude will be

$$U = 0$$

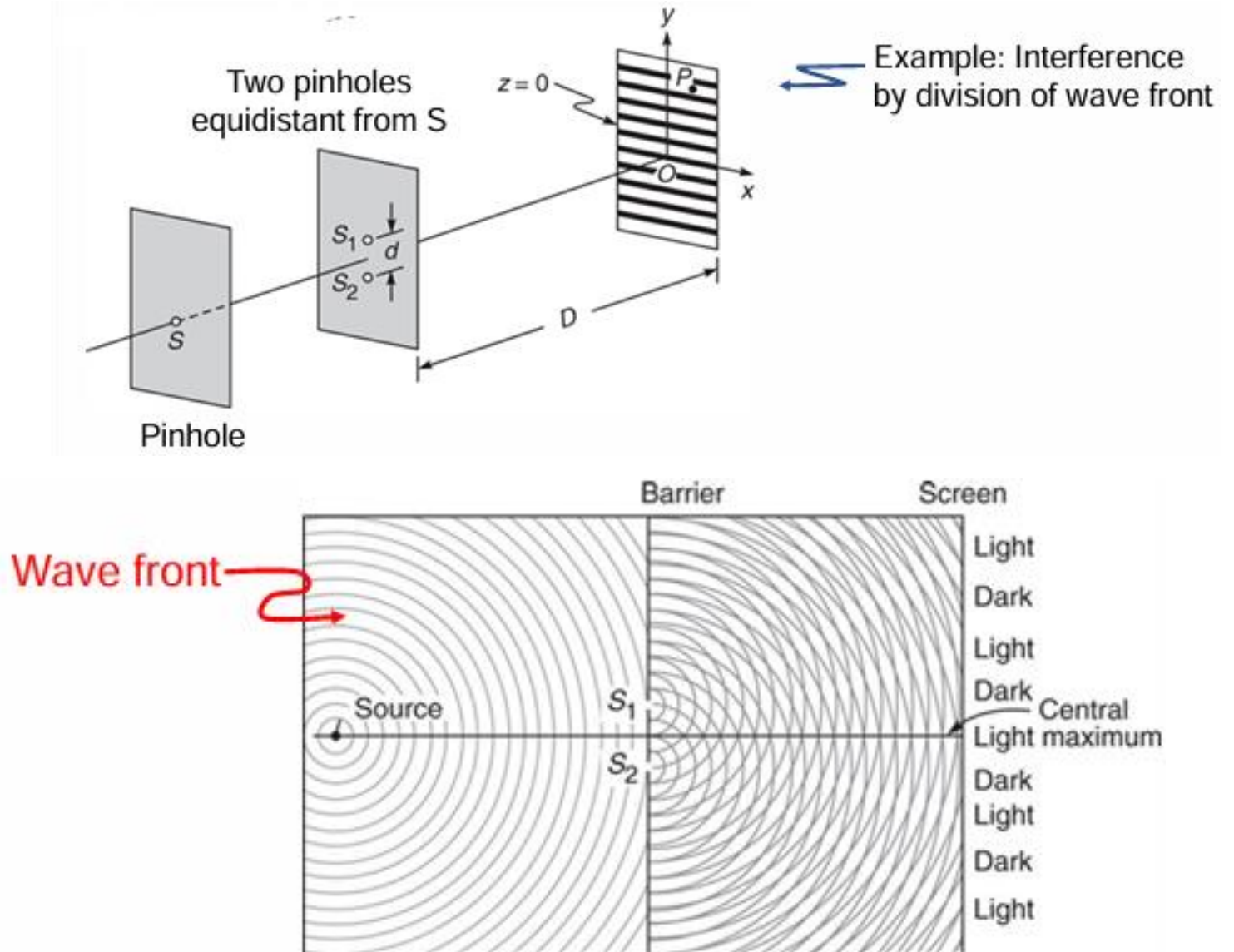
General statement: Light beams from two independent sources are incoherent

Young's double hole experiment in 1801 :

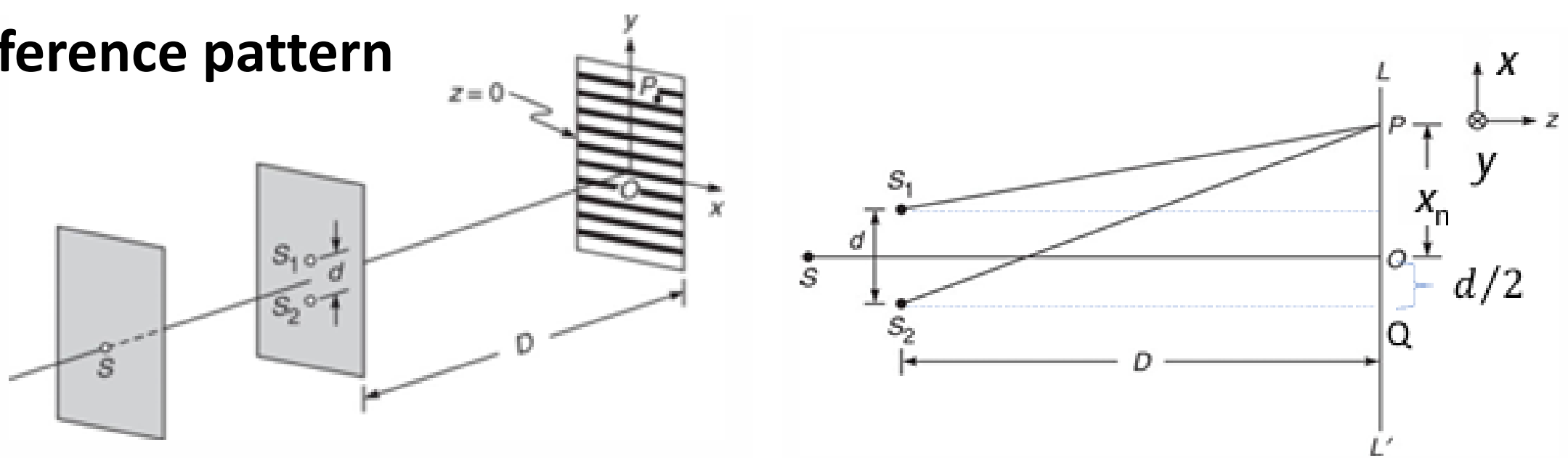
Young showed a simple method to lock the phase relationship between two sources. The trick lies in the division of a single wavefront into two.

These two split wavefronts act as if they emanate from sources having constant phase relationship.

When these two waves are allowed to interfere, a stationary wavefront is observed



Interference pattern



The phase difference between waves can change if the waves travel paths of different length
 For a max at P, we must have appropriate path difference between the two light waves from $S_{1,2}$

$$S_2P - S_1P = m\lambda_o ; m = 0, 1, 2, \dots \text{ (For bright fringes)}$$

Thus,

$$\begin{aligned} (S_2P)^2 - (S_1P)^2 &= D^2 + \left(x_n + \frac{d}{2}\right)^2 - D^2 + \left(x_n - \frac{d}{2}\right)^2 = 4x_n \frac{d}{2} \\ \Rightarrow (S_2P + S_1P) \times (S_2P - S_1P) &= 2x_n d \quad \dots (S_2P + S_1P \cong 2D) \\ \Rightarrow (S_2P - S_1P) &= \frac{2x_n d}{2D} \end{aligned}$$

$$\Rightarrow \underbrace{(S_2P - S_1P)}_{\Delta} = \frac{x_n d}{D}$$

$$\Rightarrow x_n = \frac{\Delta D}{d} = \frac{n\lambda D}{d}$$

For interference pattern
Path length difference = $n\lambda$
where n is order.

Thus distance (β) between two consecutive bright or dark fringes :

$$\Rightarrow \text{Fringe width : } x_{n+1} - x_n = (n+1 - n) \frac{\lambda D}{d}$$

$$\beta = \frac{\lambda D}{d}$$

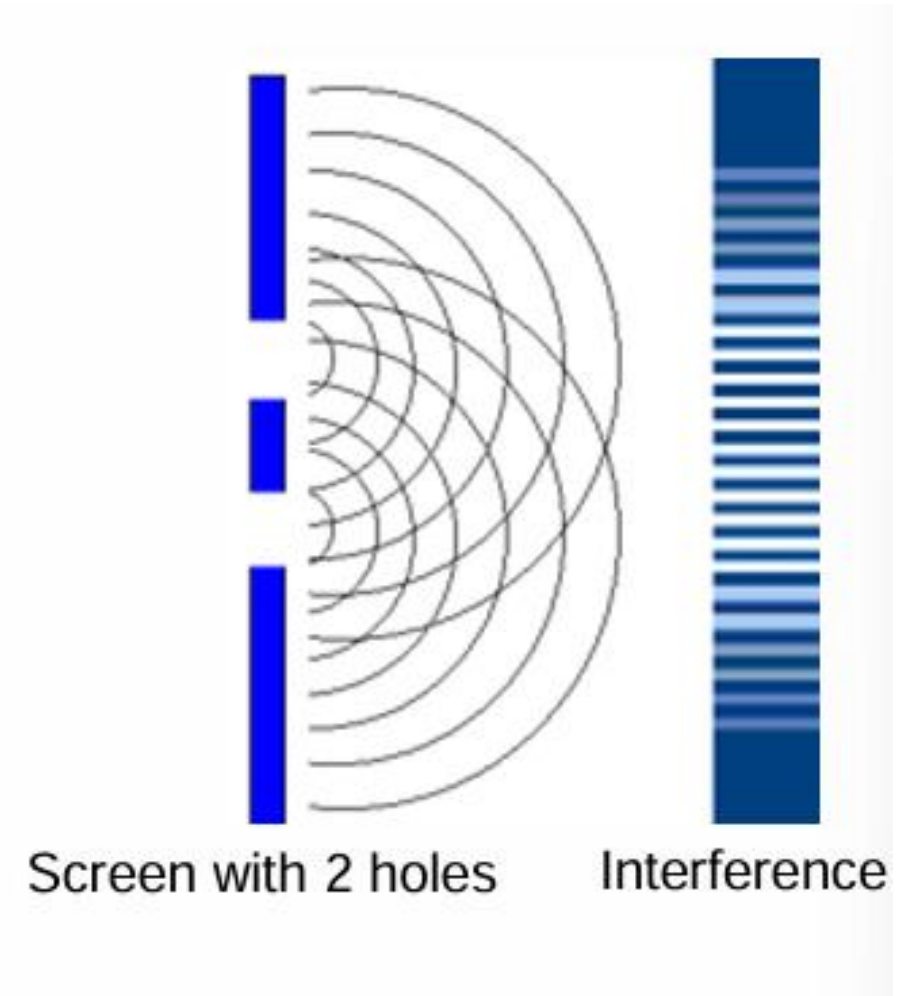
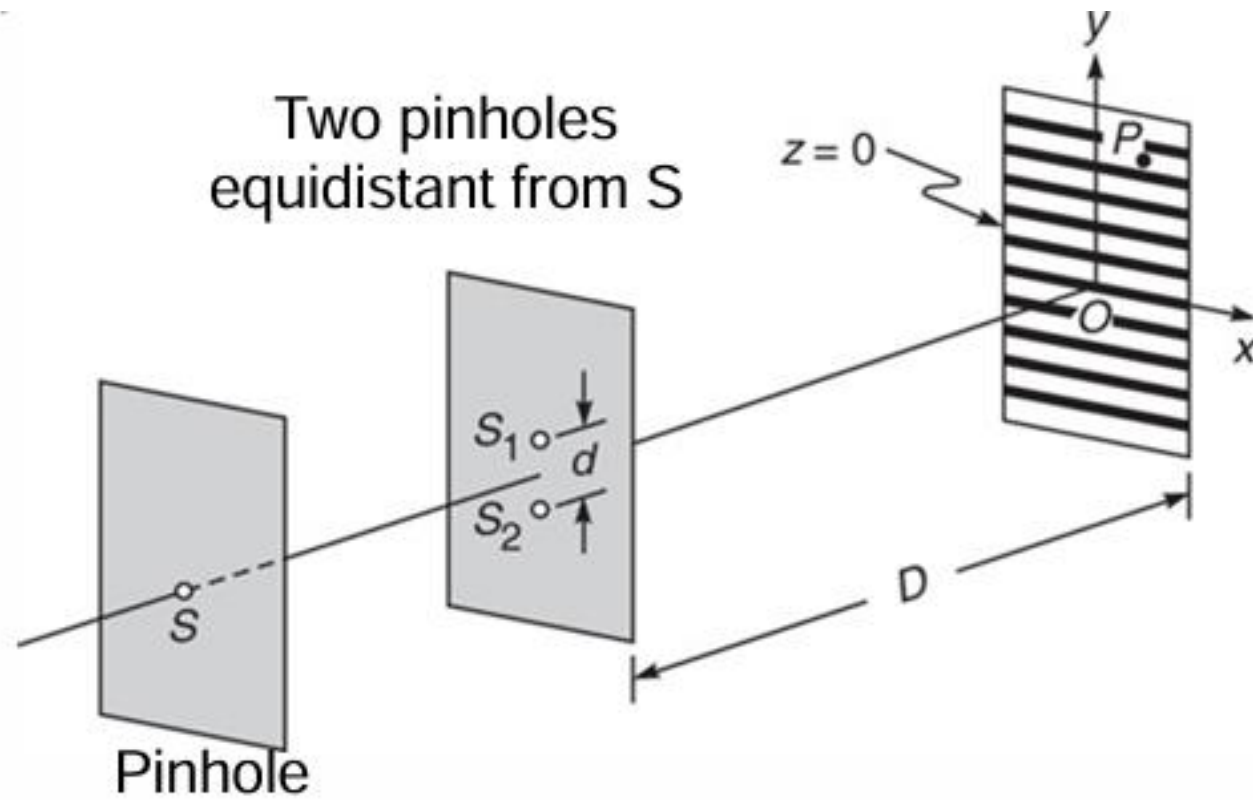
$\Rightarrow$ Bright and dark fringes will be equally spaced

Shape of the Interference fringes: Hyperbolic (Derive yourself).

Intensity Distribution: $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$ (Derive yourself)

Displacement of fringes = $\frac{D(n-1)t}{d}$ Derive yourself (These will be part of assignments)

Interference by division of wavefront :



Thanks